RIDE PULSE - A BUS TRACKING SYSTEM

REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF

FIVE YEARS INTEGRATED MASTERS (B.Sc. & M.Sc.)
IN
SOFTWARE DEVELOPMENT

By

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UNDER THE GUIDANCE

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DECLARATION

I ***Jogesh Upadhyaya***, Roll No ***IS-241-844-0014***, a Five Years Integrated Masters (B.Sc. & M.Sc.) in Software Development student of the department of Information Technology, Gauhati University hereby declares that I have compiled this report reflecting all my works during the semester long full time project as part of my curriculum.

I declare that I have included the descriptions etc. of my project work, and nothing has been copied/replicated from other's work. The facts, figures, analysis, results, claims etc. depicted in my thesis are all related to my full time project work.

I also declare that the same report or any substantial portion of this report has not been submitted anywhere else as part of any requirements for any degree/diploma etc.

Jogesh Upadhyaya
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CERTIFICATE

This is to certify that **Jogesh Upadhyaya,** bearing Roll No: ***IS-241-844-0014*** has carried out the project work ***Ride Pulse*** under my supervision and has compiled this report reflecting the candidate's work in the semester long project. The candidate did this project full time during the whole semester under my supervision, and the analysis, results, claims etc. are all related to his/her studies and works during the semester.

I recommend submission of this project report as a part for partial fulfillment of the requirements for the degree of Five Years Integrated Masters (B.Sc. & M.Sc.) in Software Development of Gauhati University.

Bijit Barman
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Jogesh Upadhyaya

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ABSTRACT

Public transportation remains one of the most widely used modes of travel in urban areas. However, passengers frequently face uncertainty regarding the arrival time and current location of buses. This often leads to long waiting times and inconvenience, especially during peak hours. The absence of a simple and accessible bus tracking mechanism can significantly affect the commuting experience.

RidePulse is a web-based real-time city bus tracking system developed to address this issue. The system allows bus drivers to share their live location using their mobile devices while passengers can monitor the current position of buses through an interactive web interface. The project combines modern web technologies such as HTML, CSS, JavaScript, Leaflet Maps, Firebase Realtime Database, and GPS-based location services to create a functional prototype of a smart transportation tracking platform.

The application consists of multiple modules including route viewing, driver registration, driver login, driver dashboard, live GPS tracking, and passenger tracking. The driver's location is continuously synchronised with Firebase, enabling passengers to receive updated location information from any device connected to the internet.

The project demonstrates the practical implementation of cloud computing, real-time databases, geolocation services, and web application development. The developed system aims to improve passenger convenience, reduce uncertainty, and showcase how technology can contribute to better urban transportation services.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Public transportation is one of the most important services in any city. It helps students, office workers, business people, and other commuters travel from one place to another in an affordable manner. In a growing city like Guwahati, buses remain one of the most commonly used modes of transport for daily travel. Every day, thousands of passengers depend on city buses to reach educational institutions, workplaces, markets, hospitals, and other destinations.

Despite the importance of public transportation, passengers often face difficulties while using bus services. One of the most common problems is the lack of information regarding the current location of buses. People usually have to wait at bus stops without knowing when a particular bus will arrive. During bad weather conditions or peak traffic hours, this uncertainty becomes even more frustrating. In many situations, passengers miss buses simply because they are unaware of the bus's current position.

With the advancement of modern technology, real-time tracking systems have become increasingly popular. GPS-enabled devices and cloud-based applications have made it possible to monitor the location of vehicles in real time. Many developed cities around the world use smart transportation systems to provide accurate and up-to-date information to passengers. Such systems help improve convenience, reduce waiting time, and increase confidence in public transportation services.

Keeping these challenges and opportunities in mind, RidePulse has been developed as a real-time city bus tracking system. The project aims to provide passengers with live information regarding bus locations through a simple and user-friendly web application. At the same time, it allows drivers to share their live GPS location through a dedicated driver portal. The information is synchronised through a cloud-based database, making it accessible to passengers from different devices.

RidePulse combines web technologies, cloud computing, GPS services, and interactive mapping tools to create a practical solution for public transportation tracking. The system demonstrates how modern software development techniques can be used to solve real-world transportation problems.

1.2 Problem Statement

One of the major problems faced by public transport users is the inability to know the exact location of buses in real time. In Guwahati, passengers often wait at bus stops without any reliable information about when the bus will arrive. Existing communication methods such as asking other passengers, contacting drivers, or relying on assumptions are inefficient and unreliable.

The absence of a centralised bus tracking platform creates several difficulties for passengers. It leads to unnecessary waiting, uncertainty, inconvenience, and poor travel planning. Students and working professionals are particularly affected because delays in transportation may impact their academic or professional schedules.

Another challenge is that many existing tracking solutions are either unavailable for local city buses or require expensive infrastructure and dedicated hardware. Small transportation operators may find it difficult to implement complex tracking systems due to financial and technical constraints.

Therefore, there is a need for a simple, affordable, and accessible bus tracking solution that can provide real-time information to passengers while remaining easy to use for drivers and transport operators.

1.3 Objectives of the Project

The primary objective of RidePulse is to develop a web-based real-time bus tracking system that improves the commuting experience of passengers.

The specific objectives of the project are as follows:

- To design and develop a user-friendly web application for bus tracking.
- To enable drivers to share their live GPS location through a dedicated driver interface.
- To provide passengers with real-time information regarding the location of buses.
- To integrate cloud-based synchronisation for sharing location data across multiple devices.
- To reduce uncertainty and waiting time for passengers.
- To demonstrate the practical application of GPS technology and cloud databases.
- To provide an affordable and scalable solution that can be extended in the future.

1.4 Scope of the Project

The scope of RidePulse is focused on developing a prototype system capable of demonstrating real-time bus tracking functionality. The project provides a foundation for future expansion into a larger transportation management platform.

The current implementation includes a passenger interface for viewing routes and tracking buses, a driver interface for registration and login, GPS-based location sharing, cloud synchronisation using Firebase Realtime Database, and map visualisation using Leaflet Maps.

The system can be accessed through modern web browsers on both desktop and mobile devices. Since it is a web-based solution, no additional software installation is required for users.

Although the current version focuses on a limited number of routes, the architecture has been designed in a way that allows future integration of multiple buses, additional routes, arrival time prediction, administrative controls, and mobile applications.

The project serves as a proof-of-concept demonstrating how technology can be used to improve public transportation services in cities like Guwahati.

1.5 Significance of the Project

The significance of RidePulse lies in its ability to address a practical problem faced by a large number of commuters. By providing real-time location information, the system helps passengers make better travel decisions and reduces unnecessary waiting time.

The project also demonstrates the integration of several modern technologies including web development, geolocation services, cloud databases, and interactive maps. This makes it not only a useful transportation application but also an educational project that showcases important software development concepts.

From a social perspective, the system contributes towards smarter urban transportation. Better access to transport information can encourage more people to use public transportation, which may help reduce traffic congestion and promote sustainable mobility.

Furthermore, RidePulse can serve as a starting point for future research and development in intelligent transportation systems. The project has the potential to be expanded into a full-scale city-wide bus tracking platform capable of serving a larger number of users and routes.

1.6 Organisation of the Report

This report has been organised into six chapters.

Chapter 1 introduces the project, discusses the problem statement, objectives, scope, significance, and overall structure of the report.

Chapter 2 presents the background study and literature review related to transportation systems, GPS tracking technologies, and existing tracking solutions.

Chapter 3 explains the methodology adopted for the project, including requirement analysis, system design, development approach, and system architecture.

Chapter 4 describes the implementation details of RidePulse, including various modules, technologies used, Firebase integration, and deployment.

Chapter 5 discusses the results obtained from testing and evaluating the system.

Chapter 6 presents the conclusion of the project and suggests possible future enhancements.

CHAPTER 2

BACKGROUND AND LITERATURE REVIEW

2.1 Introduction

Transportation plays a crucial role in the development of modern cities. Efficient transportation systems contribute to economic growth, social development, and improved quality of life. Public transportation, in particular, serves as an affordable and accessible mode of travel for a large section of the population. Buses are among the most widely used forms of public transportation due to their flexibility, lower operational costs, and ability to connect various parts of a city.

With the rapid growth of urban populations, transportation systems face several challenges such as traffic congestion, delays, route management, and passenger dissatisfaction. One of the major concerns of public transport users is the lack of real-time information regarding vehicle locations and arrival times. Modern technology has introduced several solutions to address these issues through GPS tracking, cloud computing, and mobile applications.

This chapter presents the background information and literature review related to transportation systems, GPS technology, cloud-based tracking solutions, and existing applications relevant to the RidePulse project.

2.2 Public Transportation Systems

Public transportation refers to transport services that are available for use by the general public. These services include buses, trains, metro systems, ferries, and other forms of shared transportation. Public transport helps reduce traffic congestion, minimises fuel consumption, and provides affordable mobility to citizens.

In cities like Guwahati, buses form an important component of the transportation network. Thousands of passengers rely on city buses every day for commuting to educational institutions, offices, markets, hospitals, and other destinations. Despite their importance, many bus services continue to operate without advanced passenger information systems.

Traditional bus transportation systems often lack mechanisms for providing real-time information to passengers. As a result, passengers are forced to wait at bus stops without knowing the exact location of buses or expected arrival times. This creates uncertainty and negatively impacts the overall travel experience.

The growing demand for smarter transportation services has encouraged the adoption of digital technologies capable of providing real-time tracking and information-sharing capabilities.

2.3 Global Trends in Smart Transportation

The concept of smart transportation has gained significant attention in recent years. Smart transportation systems use information technology, communication networks, sensors, cloud computing, and GPS devices to improve the efficiency and reliability of transport services.

Many developed countries have implemented Intelligent Transportation Systems (ITS) that allow passengers to access live transportation information through websites and mobile applications. These systems provide features such as vehicle tracking, estimated arrival times, route planning, and traffic monitoring.

Applications such as Google Maps, public transit tracking platforms, and ride-sharing services have demonstrated the effectiveness of location-based technologies. By providing accurate and timely information, these systems help users make informed travel decisions and improve overall transportation efficiency.

The increasing availability of smartphones and internet connectivity has further accelerated the adoption of smart transportation solutions worldwide.

2.4 Global Positioning System (GPS)

The Global Positioning System (GPS) is a satellite-based navigation system that enables users to determine their geographic location anywhere on Earth. GPS technology has become one of the most important components of modern tracking systems.

A GPS receiver communicates with multiple satellites orbiting the Earth to calculate its position based on signals received from those satellites. The calculated position is typically represented using latitude and longitude coordinates.

GPS technology offers several advantages:

- Accurate location tracking.
- Real-time position updates.
- Wide geographic coverage.
- Integration with mobile devices.
- Support for navigation and mapping applications.

Because of these advantages, GPS has become the preferred technology for vehicle tracking, fleet management, logistics monitoring, and transportation applications.

In the RidePulse project, GPS technology is used to obtain the live location of bus drivers and transmit that information to passengers through a web-based interface.

2.5 Cloud Computing and Real-Time Databases

Cloud computing has transformed the way data is stored, processed, and shared. Instead of maintaining dedicated servers, developers can use cloud platforms to manage data and synchronise information across multiple devices.

One of the most important developments in cloud computing is the emergence of real-time databases. Unlike traditional databases, real-time databases instantly synchronise updates among connected users. This makes them highly suitable for applications that require continuous information sharing.

Firebase Realtime Database is a cloud-hosted NoSQL database developed by Google. It enables data synchronisation in real time and supports web, mobile, and desktop applications.

Some key advantages of Firebase Realtime Database include:

- Real-time data synchronisation.
- Cloud-based accessibility.
- Easy integration with web applications.
- Reduced backend development complexity.
- Support for cross-device communication.

In RidePulse, Firebase Realtime Database acts as the communication bridge between drivers and passengers. Location updates sent by drivers are immediately reflected on the passenger tracking interface.

2.6 Web-Based Tracking Systems

Web-based tracking systems have become increasingly popular because they can be accessed directly through web browsers without requiring software installation. Such systems are platform-independent and can be used on desktops, laptops, tablets, and smartphones.

A typical web-based tracking system consists of the following components:

- User Interface
- Location Services
- Database Management
- Communication Layer
- Visualisation Tools

Modern web technologies such as HTML, CSS, and JavaScript enable developers to create interactive and responsive tracking applications. The integration of mapping libraries such as Leaflet allows users to visualise geographical locations on interactive maps.

Web-based tracking systems are widely used in transportation, logistics, delivery services, ride-sharing platforms, and fleet management applications.

2.7 Review of Existing Solutions

Several transportation tracking systems are available globally. Applications such as Google Maps provide live transit information in many cities. Ride-sharing platforms such as Uber and Ola also use real-time location tracking to improve user experience.

However, these solutions are often designed for large-scale transportation networks and may not be directly applicable to local city bus services. Many local transportation operators lack the infrastructure and financial resources required to implement sophisticated tracking platforms.

Furthermore, some existing systems depend on dedicated hardware devices, increasing implementation costs. Small and medium-sized transportation providers may find such solutions difficult to adopt.

These limitations highlight the need for simple, affordable, and scalable solutions that can be implemented using commonly available technologies.

2.8 Need for RidePulse

The review of existing systems indicates that there is a growing need for accessible transportation tracking solutions. Passengers require accurate information regarding bus locations, while transportation operators need affordable technologies that can be implemented without significant investment.

RidePulse addresses this need by utilising widely available technologies such as smartphones, GPS services, cloud databases, and web browsers. Instead of relying on expensive hardware, the system uses the driver's mobile device to share location information.

The project demonstrates how modern web technologies can be combined to create a practical transportation solution suitable for local conditions. It serves as a prototype for future smart transportation initiatives and provides a foundation for additional enhancements such as arrival time prediction, route analytics, and multi-bus management.

2.9 Chapter Summary

This chapter discussed the background concepts and existing technologies relevant to the development of RidePulse. Topics including public transportation systems, GPS technology, cloud computing, Firebase Realtime Database, web-based tracking systems, and existing transportation solutions were examined. The review highlighted the importance of real-time tracking and established the need for a system such as RidePulse. The next chapter describes the methodology adopted for the development of the project and presents the overall system architecture.

CHAPTER 3

METHODOLOGY AND SYSTEM ARCHITECTURE

3.1 Introduction

The successful development of any software system requires a structured approach that guides the project from idea generation to final implementation. Methodology refers to the process and techniques used during the development of a project. It helps developers organise tasks, identify requirements, design system components, implement features, and evaluate the final product.

RidePulse was developed using a practical and iterative development approach. The project began with identifying the transportation problems faced by passengers and gradually evolved into a web-based real-time tracking system. Multiple technologies were integrated together to achieve the final objective of providing live bus tracking through a web interface.

This chapter explains the methodology adopted during development and presents the architecture of the RidePulse system.

3.2 Requirement Analysis

Requirement analysis was the first stage of development. The objective of this stage was to understand the needs of both passengers and drivers and identify the features required in the system.

The following requirements were identified:

Passenger Requirements:

- Ability to view available bus routes.
- Ability to track bus location in real time.
- Access to the system through a web browser.
- Simple and user-friendly interface.

Driver Requirements:

- Driver registration facility.
- Driver login functionality.
- Ability to share live location.
- Secure access to the tracking system.

System Requirements:

- Real-time location synchronisation.
- Cloud-based data storage.
- Interactive map visualisation.
- Cross-device accessibility.

The analysis of these requirements formed the foundation for the overall system design.

3.3 Development Methodology

RidePulse was developed using an iterative development methodology. Instead of building the entire system at once, the project was divided into smaller modules and developed step by step.

The development process followed the following stages:

1. Requirement Analysis

The transportation problem was studied and project requirements were identified.

2. System Design

User interfaces, navigation structure, and system workflow were planned before coding.

3. Frontend Development

HTML, CSS, and JavaScript were used to develop the user interface.

4. Map Integration

Leaflet Maps were integrated to provide geographical visualisation of bus locations.

5. Database Integration

Firebase Realtime Database was integrated to enable cloud-based synchronisation.

6. GPS Integration

Browser geolocation services were implemented to capture driver location data.

7. Testing and Debugging

The system was tested across multiple devices and browsers.

8. Deployment

The final application was deployed using Vercel for public accessibility.

This iterative approach allowed continuous improvements and easier debugging during development.

3.4 System Architecture

The RidePulse system follows a client-cloud-client architecture.

The architecture consists of five major components:

1. Driver Device

The driver uses a smartphone or GPS-enabled device to access the Driver Panel. The browser obtains the driver's location using the Geolocation API.

2. GPS Service

The browser collects latitude and longitude coordinates from the device's GPS hardware.

3. Firebase Realtime Database

The location data is transmitted to Firebase Realtime Database, which acts as a cloud communication layer.

4. Passenger Interface

Passengers access the Track Page through their web browser. The application retrieves location data from Firebase.

5. Interactive Map

Leaflet Maps is used to display the live location of buses on an interactive map interface. The mapping component provides a visual representation of the bus position, allowing passengers to monitor the movement of the vehicle in real time. The map updates automatically whenever new location data is received from the database, ensuring that passengers always have access to the most recent location information.

The overall architecture of the RidePulse system consists of five major components: the driver device, GPS location services, Firebase Realtime Database, the RidePulse web application, and the passenger device. The driver's device continuously collects location coordinates using the Geolocation API. These coordinates are transmitted to Firebase Realtime Database, which acts as the central communication layer of the system. The RidePulse web application retrieves the latest location information from Firebase and displays it to passengers through an interactive map interface.

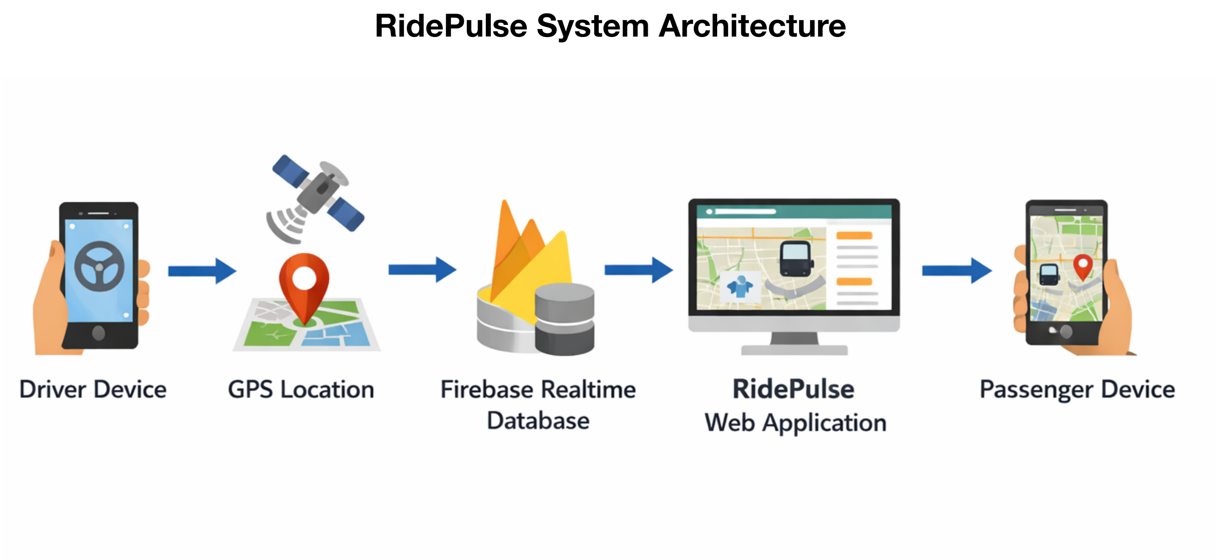

Figure 3.1 Overall System Architecture of RidePulse.

The selected architecture enables real-time communication between drivers and passengers without requiring a dedicated backend server. By utilising cloud-based synchronisation through Firebase, the system ensures that location updates are immediately reflected across all connected devices, thereby providing an efficient and scalable solution for real-time bus tracking.

3.5 System Workflow

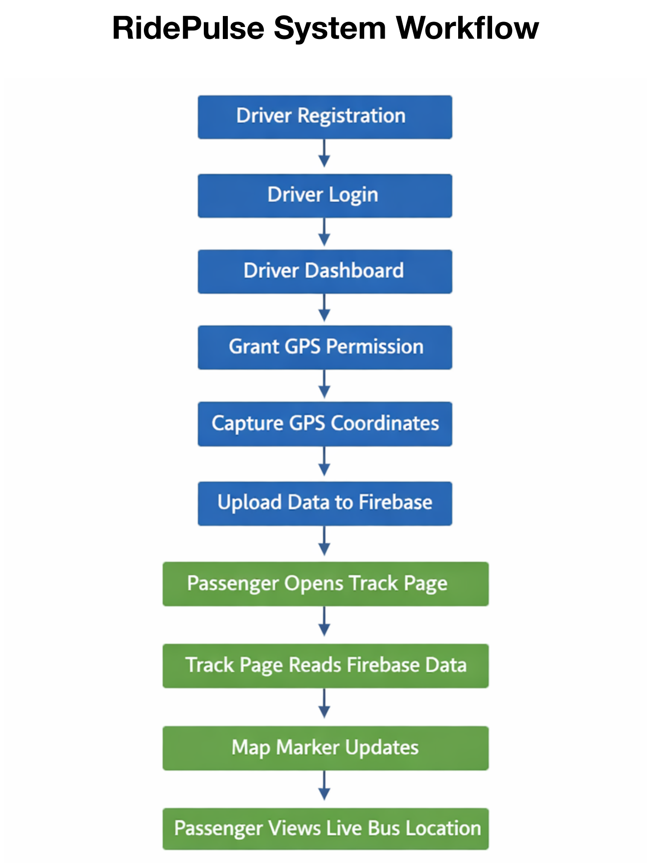

Figure 3.2 Workflow of the RidePulse Real-Time Bus Tracking System

The workflow of RidePulse describes the sequence of operations performed by the system. The system begins when a driver registers and logs into the application. After authentication, the driver enables live tracking, allowing GPS coordinates to be collected and uploaded to Firebase Realtime Database. Passengers access the Track Bus page, where real-time location updates are retrieved from Firebase and displayed on an interactive map.

3.6 Technologies Used

Several modern technologies were used during the development of RidePulse.

HTML

HTML was used to create the structure of web pages including forms, buttons, navigation bars, and content sections.

CSS

CSS was used to design the user interface and improve visual appearance. Responsive design techniques were implemented to ensure compatibility across different devices.

JavaScript

JavaScript was used to implement system logic, event handling, user interactions, GPS integration, and Firebase communication.

Leaflet Maps

Leaflet is an open-source JavaScript library used for displaying interactive maps. It was integrated to visualise bus locations.

Firebase Realtime Database
Firebase Realtime Database was used to store and synchronise live location data between devices.

Geolocation API

The Geolocation API was used to obtain latitude and longitude coordinates from the driver's device.

Vercel

Vercel was used to deploy and host the final application on the internet.

3.7 Database Structure

RidePulse utilises Firebase Realtime Database as a cloud-based NoSQL database for storing and synchronising live bus location information. Firebase was selected because it provides real-time data synchronisation, enabling seamless communication between the driver and passenger interfaces.

The primary database node used in the system is named liveBus. This node stores the current location and identification details of the active bus being tracked.

Table 3.1: Structure of the liveBus Node

Field Name	Data Type	Description
Bus	String	Stores the identification number of the bus
Lat	Decimal	Stores the current latitude coordinate of the bus
Lng	Decimal	Stores the current longitude coordinate of the bus

A sample record stored in the database is shown below.

Table 3.2: Sample Database Record

Bus	Lat	Lng
101	26.1825	91.7500

The database is continuously updated whenever the driver's location changes. Location coordinates obtained through the Geolocation API are transmitted to Firebase Realtime Database in real time. As soon as new coordinates are uploaded, Firebase automatically synchronises the updated information across connected devices. This enables passengers to view the latest bus location without manually refreshing the application.

The use of Firebase Realtime Database eliminates the need for a dedicated backend server while ensuring fast and reliable communication between drivers and passengers.

3.8 Data Flow Mechanism

The RidePulse system uses Firebase Realtime Database as an intermediary communication layer between drivers and passengers. When a driver enables live tracking through the Driver Dashboard, the browser obtains GPS coordinates using the Geolocation API. These coordinates are transmitted to Firebase Realtime Database and stored in real time.

The Passenger Tracking Module continuously listens for updates from Firebase. Whenever the location data changes, the updated coordinates are automatically retrieved and displayed on the tracking interface. This process eliminates the need for manual page refreshes and ensures that passengers receive the latest location information.

The use of Firebase as a cloud-based synchronisation platform enables seamless communication between multiple devices connected to the internet. This mechanism forms the foundation of the real-time tracking capability provided by RidePulse.

3.9 Advantages of the Selected Architecture

The architecture selected for RidePulse offers several advantages.

- Real-time synchronisation.
- Low implementation cost.
- No dedicated server requirement.
- Cross-device compatibility.
- Easy deployment and maintenance.
- Scalable for future expansion.
- Suitable for web-based transportation systems.

These advantages make the architecture suitable for the objectives of the project.

3.10 Chapter Summary

This chapter discussed the methodology adopted during the development of RidePulse and explained the overall system architecture. Requirement analysis, development methodology, workflow, database structure, and technologies used in the project were presented. The architecture demonstrates how GPS services, Firebase Realtime Database, and web technologies work together to provide real-time bus tracking. The next chapter describes the implementation details of various modules developed within the RidePulse system.

CHAPTER 4

PROPOSED SOLUTION AND IMPLEMENTATION

4.1 Introduction

The primary objective of RidePulse is to address the problem of uncertainty faced by passengers while waiting for city buses. Traditional public transportation systems often lack real-time tracking capabilities, making it difficult for passengers to know the current location of buses. As a result, passengers spend considerable time waiting at bus stops without accurate information regarding bus arrival.

To overcome this problem, RidePulse has been developed as a web-based real-time bus tracking system. The proposed solution utilises GPS technology, cloud-based data synchronisation, and interactive maps to provide live location information to passengers. The system allows drivers to share their location through a dedicated driver interface while passengers can monitor the movement of buses through a tracking interface.

The implementation of RidePulse focuses on simplicity, accessibility, and affordability. The system is designed to operate using commonly available smartphones and internet connectivity without requiring specialised hardware.

4.2 Proposed Solution

The proposed solution consists of two primary user groups:

1. Drivers

Drivers are responsible for sharing the real-time location of buses. A dedicated Driver Access module allows drivers to register, log in, and activate live location sharing.

2. Passengers

Passengers access the RidePulse website to view available routes and track buses in real time. The system displays the current bus location on an interactive map.

The proposed solution eliminates the need for passengers to rely on assumptions or manual communication regarding bus arrivals. Instead, location updates are automatically synchronised through a cloud-based database and presented visually through the web application.

The solution combines the following technologies:

- GPS Location Tracking
- Firebase Realtime Database
- Web-Based User Interface
- Interactive Mapping System
- Cloud Synchronisation

This combination provides a practical and scalable approach to public transportation tracking.

4.3 Home Page Implementation

The Home Page serves as the entry point of the RidePulse application. It provides users with a brief overview of the project and allows navigation to other modules of the system.

The Home Page contains:

- Project branding and title.
- Navigation bar.
- Introduction to the RidePulse system.
- Access to route information.
- Access to the Driver Portal.

Home Page

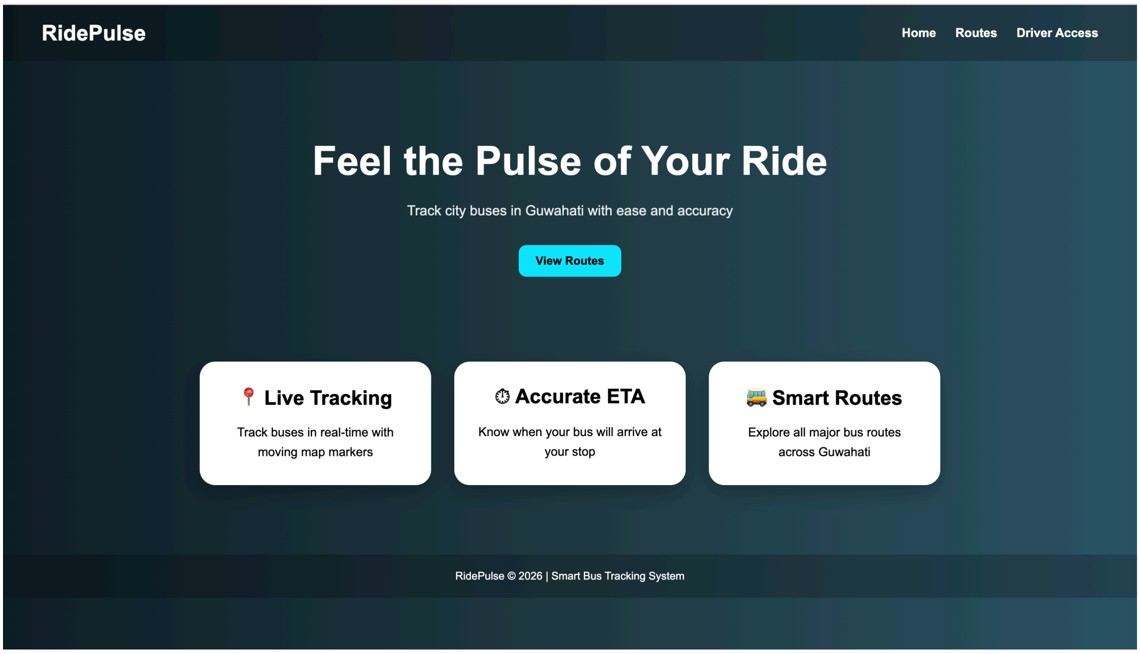

Figure 4.1 Homepage Interface of the RidePulse System

The page was designed using HTML and CSS to provide a clean and responsive user interface. Particular attention was given to mobile compatibility since many users are expected to access the system using smartphones.

4.4 Routes Module Implementation

The Routes Module displays available bus routes supported by the system. This module provides passengers with information regarding the routes that can be tracked through RidePulse.

The implementation includes:

- Route cards.
- Route descriptions.
- Track Bus buttons.
- Status indicators.

Routes Module

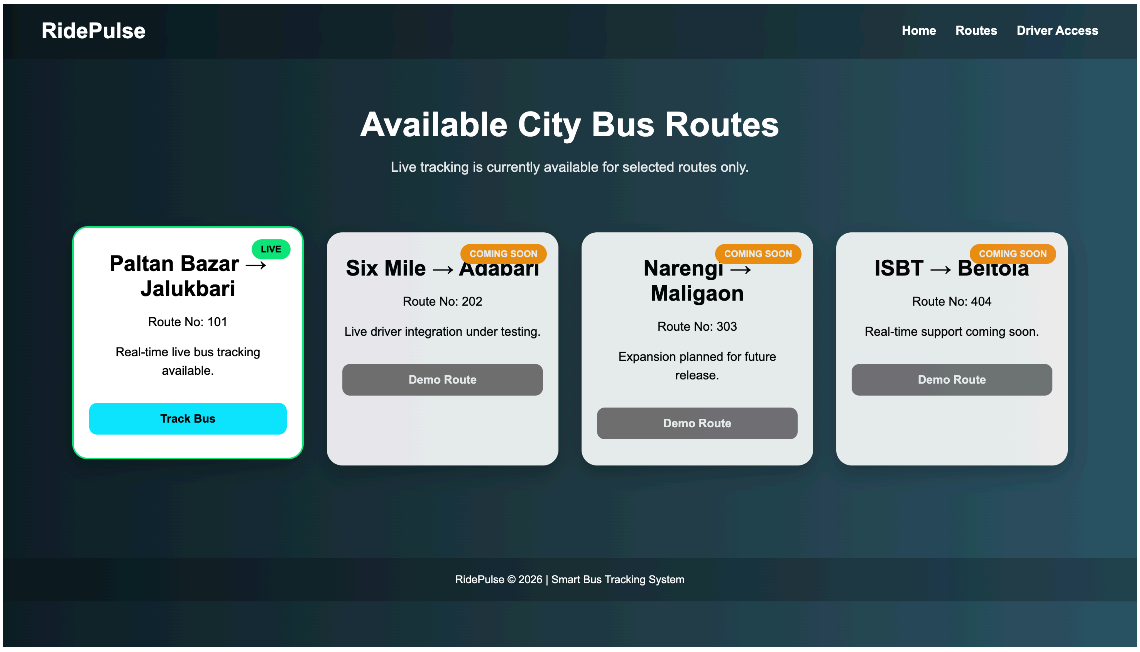

Figure 4.2 Routes Interface Showing Available Bus Routes and Live Tracking Availability

The route information is displayed using responsive card layouts that automatically adjust according to screen size.

The Track Bus button redirects users to the live tracking interface.

4.5 Driver Registration Module

Driver Login & Registration Module

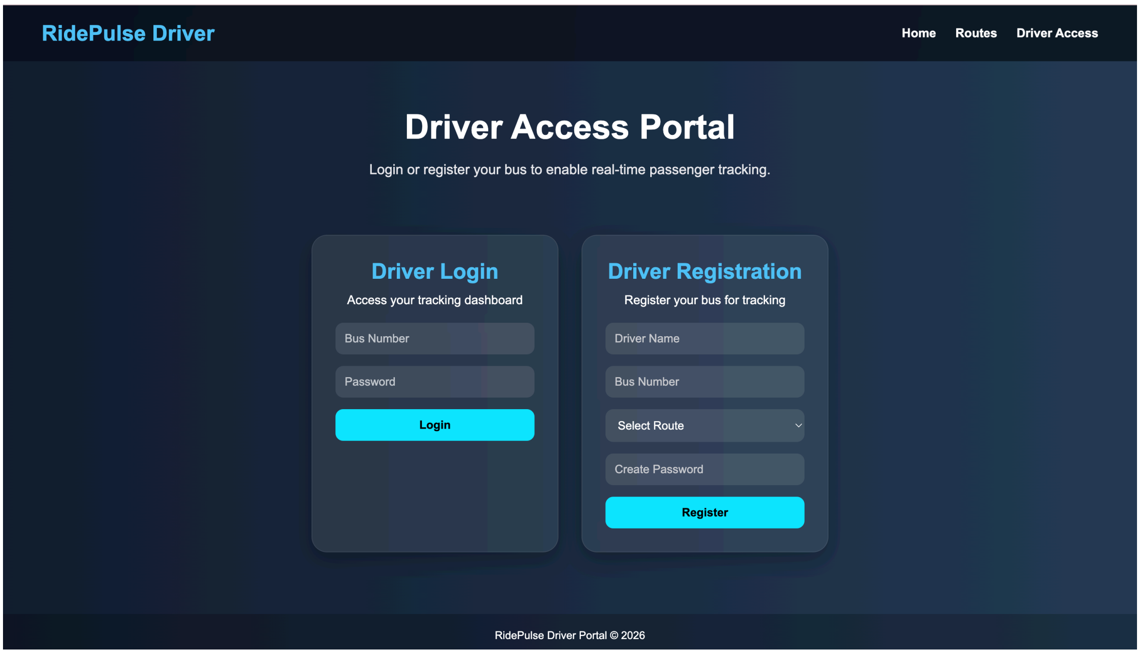

Figure 4.3 Driver Authentication Interface Showing Login and Registration Modules

The Driver Registration Module allows drivers to create an account within the system. The registration form collects basic information required for identification and tracking.

The information collected includes:

Table 4.1: Driver Registration Information

Field	Purpose
Driver Name	Identifies the driver
Bus Number	Identifies the bus
Route	Assigns a route
Password	Authentication

4.6 Driver Login Module

The Driver Login Module enables registered drivers to securely access the RidePulse tracking system. This module acts as the authentication layer of the application and ensures that only authorised drivers can access the Driver Dashboard and share live bus locations.

The login process includes:

Step 1: Enter Bus Number.

Step 2: Enter Password.

Step 3: Validate credentials.

Step 4: Redirect to Driver Dashboard.

Authentication ensures that only authorised drivers can share live location information.

The implementation was carried out using JavaScript validation and local storage mechanisms.

4.7 Driver Dashboard Implementation

The Driver Dashboard serves as the central control panel for drivers within the RidePulse system. After successful authentication, drivers are redirected to the dashboard, where they can access essential tracking functionalities and monitor their assigned route information.

The dashboard provides:

- Driver information.
- Bus information.
- Live tracking controls.
- GPS activation functionality.
- Logout functionality.

Driver Dashboard (Offline)

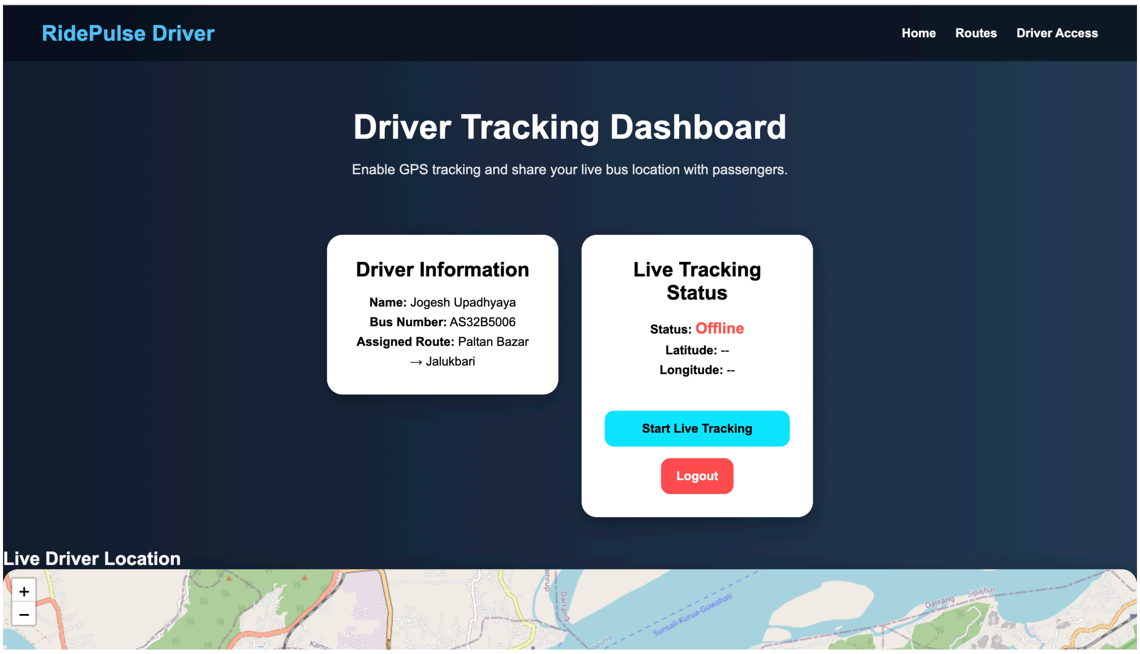

Figure 4.4 Driver Dashboard of the RidePulse System (Offline)

After successful login, drivers can activate live tracking by clicking the Start Live Tracking button.

The dashboard continuously monitors GPS coordinates and prepares location updates for synchronisation with Firebase.

4.8 GPS Integration

GPS Integration is one of the most important components of the RidePulse system, as it enables the collection of real-time bus location information. The system utilises the Geolocation API provided by modern web browsers to access the driver's current geographic coordinates.

<u>The process consists of:</u>

3. Driver grants location permission.
4. Browser retrieves GPS coordinates.

5. Latitude and longitude values are generated.
6. Coordinates are transmitted to Firebase.

Driver Dashboard (Online)

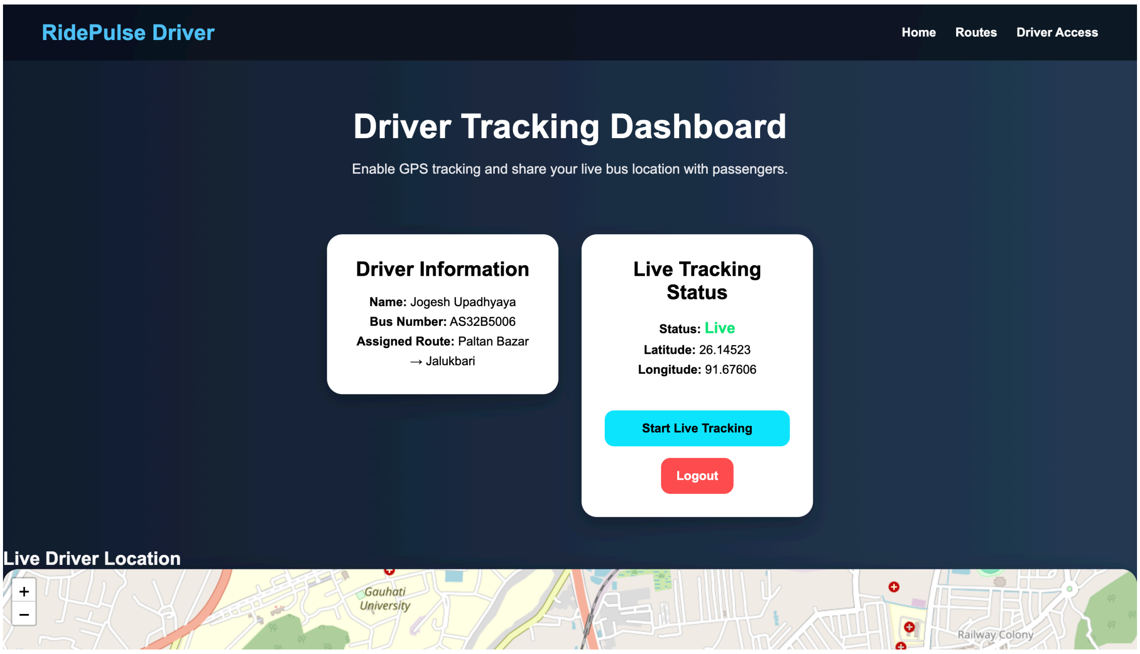

Figure 4.5 Driver Dashboard of the RidePulse System (Online)

The use of browser-based geolocation eliminates the need for external GPS hardware while maintaining acceptable accuracy.

4.9 Firebase Realtime Database Integration

Firebase Realtime Database plays a crucial role in the RidePulse system by enabling real-time communication between drivers and passengers. It acts as a cloud-based storage and synchronisation platform that continuously updates and distributes location information across connected devices.

Whenever the driver's location changes, updated coordinates are automatically transmitted to Firebase.

The database stores:

- Bus Number
- Latitude
- Longitude

Firebase Realtime Database

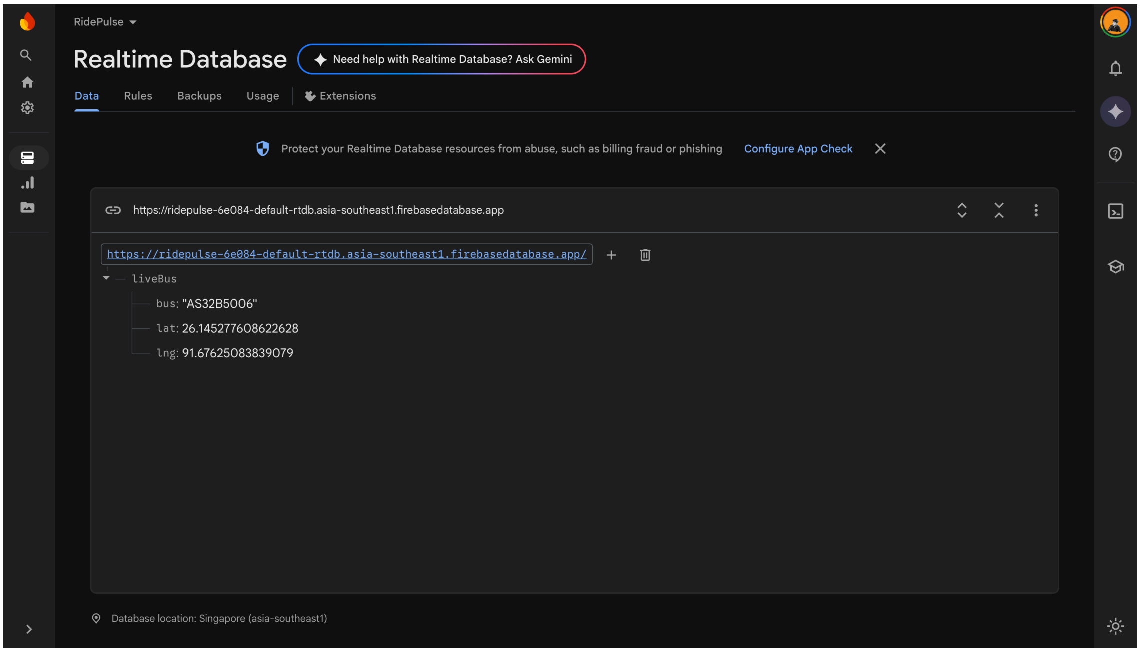

Figure 4.6 Firebase Realtime Database integrated into the RidePulse system.

The synchronisation process occurs in real time, enabling connected passengers to immediately receive updated location information.

Firebase was selected due to its:

- Real-time synchronisation capabilities.
- Cloud accessibility.
- Ease of integration.
- Scalability.

4.10 Passenger Tracking Module

The Passenger Tracking Module is the primary interface through which users monitor the live location of buses within the RidePulse system. This module is designed to provide passengers with real-time access to bus movement information, thereby improving convenience and reducing uncertainty during travel.

Passenger Tracking Module

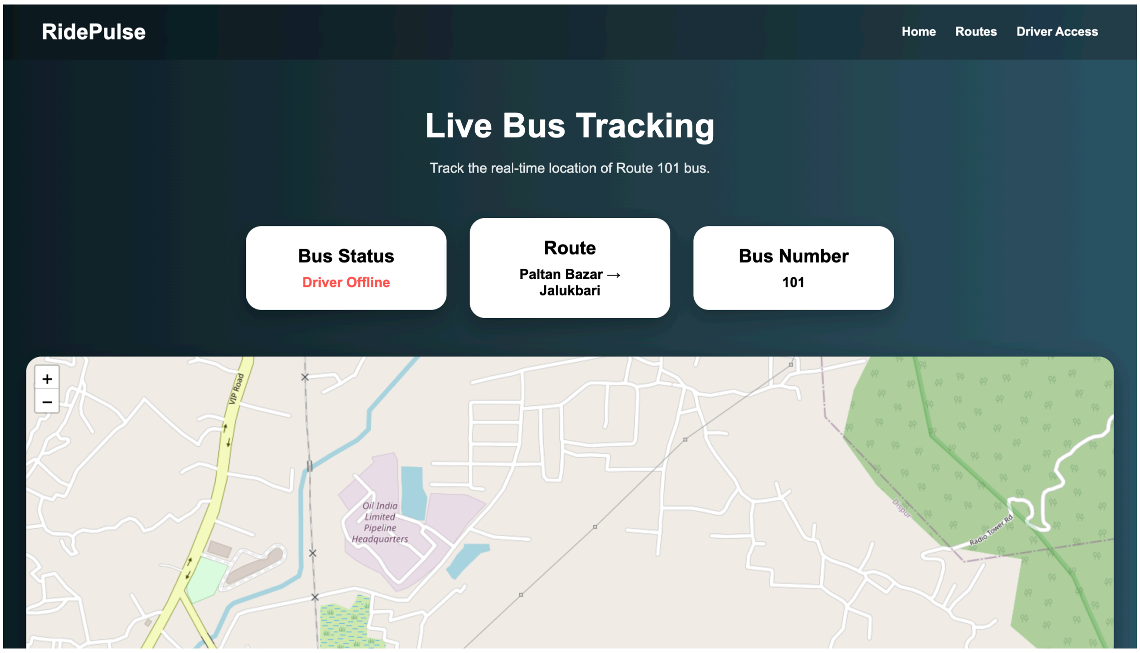

Figure 4.7 Passenger Tracking Interface, where users can monitor route information, bus status, and real-time tracking data obtained from Firebase Realtime Database.

This module retrieves live location data from Firebase and displays the information through an interactive map interface.

<u>The module includes:</u>

- Bus status display.
- Latitude display.
- Longitude display.

- Live map marker.
- Route information.

As soon as location updates are received, the map marker moves automatically to reflect the latest bus position.

4.11 Interactive Map Implementation

The map functionality was implemented using the Leaflet JavaScript library.

<u>Leaflet was selected because it is:</u>

- Open source.
- Lightweight.
- Easy to integrate.
- Compatible with OpenStreetMap.

<u>The map displays:</u>

- Current bus location.
- Marker updates.
- Real-time movement.
- Zoom controls.

The visual representation of location data improves user understanding and provides a more intuitive tracking experience.

4.12 Deployment

After successful testing and debugging, RidePulse was deployed using Vercel.

<u>The deployment process included:</u>

- Project configuration.
- Version management.
- Hosting setup.
- Public URL generation.

Deployment allows the application to be accessed through both desktop and mobile devices from any location with internet connectivity.

The deployed version demonstrates real-time communication between drivers and passengers across multiple devices.

4.13 Chapter Summary

This chapter described the proposed solution and implementation details of RidePulse. Various modules including the Home Page, Routes Module, Driver Registration, Driver Login, Driver Dashboard, GPS Integration, Firebase Integration, Passenger Tracking, Interactive Maps, and Deployment were discussed. The implementation demonstrates how web technologies and cloud services can be combined to create a practical real-time bus tracking solution. The next chapter presents the results obtained during testing and discusses the performance of the system.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

After the successful implementation of RidePulse, extensive testing was conducted to evaluate the functionality, reliability, and performance of the system. The purpose of testing was to verify whether the developed application fulfilled the objectives established during the requirement analysis stage. The testing process focused on user authentication, GPS location tracking, Firebase synchronisation, interactive map visualisation, and cross-device communication.

The results obtained during testing demonstrate the practical feasibility of the proposed solution and highlight its effectiveness in providing real-time bus tracking services.

5.2 Testing Environment

The RidePulse system was tested using commonly available hardware and software resources to evaluate its functionality, reliability, and real-time tracking capabilities. The objective of testing was to ensure that all modules of the application operated correctly under normal usage conditions and provided a consistent user experience.

Testing was performed on both desktop and mobile devices using modern web browsers and internet connectivity. Different components of the system, including driver authentication, GPS location tracking, Firebase Realtime Database synchronisation, and passenger tracking, were verified during the testing process. Particular attention was given to real-time communication between devices to ensure that location updates generated by the driver were accurately reflected on the passenger interface.

The use of multiple devices and platforms helped confirm the compatibility, responsiveness, and practical usability of the RidePulse system in a real-world environment.

Table 5.1: Testing Environment

Component	Specification
Operating System	Windows / MacOS / Android
Browser	Google Chrome
Frontend Technologies	HTML, CSS, JS
Database	Firebase Realtime Database
Mapping Library	Leaflet Maps
Hosting Platform	Vercel
Internet Connection	Mobile Data / Wi-Fi

5.3 Functional Testing

Functional testing was carried out to ensure that all modules of the RidePulse system performed according to their intended functionality and met the requirements identified during the development phase. Each feature of the application was tested individually as well as in combination with other modules to verify proper interaction between different components of the system.

The testing process included verifying driver registration, driver login, GPS location retrieval, Firebase Realtime Database synchronisation, passenger tracking, route navigation, and logout functionality. Various test scenarios were executed to confirm that the system responded correctly to user actions and handled normal operational conditions effectively.

The results obtained from functional testing demonstrated that all major modules operated as expected. User inputs were processed correctly, location data was transmitted and synchronised successfully, and real-time tracking information was accurately displayed on the passenger interface. Overall, the testing confirmed that the RidePulse system was capable of delivering its intended functionality in a reliable and consistent manner.

Table 5.2: Functional Testing Results

Test Case	Expected Result	Actual Result	Status
Home Page Loading	Homepage should load successfully	Successful	Pass
Route Display	Routes should be visible	Successful	Pass
Driver Registration	Driver should register successfully	Successful	Pass
Driver Login	Driver should login successfully	Successful	Pass
GPS Permission	Location access should be requested	Successful	Pass
Live Tracking	Bus location should update continuously	Successful	Pass
Firebase Synchronisation	Data should update in real time	Successful	Pass
Passenger Tracking	Live location should be visible	Successful	Pass
Logout Function	Session should terminate successfully	Successful	Pass

5.4 GPS Tracking Results

The GPS tracking functionality was tested using a mobile device. The driver activated live tracking through the Driver Dashboard and granted location permission through the browser.

The Geolocation API successfully retrieved latitude and longitude coordinates from the device. The coordinates were continuously updated whenever the location changed.

The obtained location data demonstrated acceptable accuracy for a prototype transportation tracking system. Small variations in accuracy were observed depending on network conditions and GPS signal strength. However, the overall performance remained satisfactory for demonstrating real-time bus tracking.

The successful retrieval and transmission of GPS coordinates confirmed the effectiveness of the location tracking module.

5.5 Firebase Synchronisation Results

Firebase Realtime Database was used as the cloud synchronisation platform. During testing, location updates generated by the driver device were transmitted to Firebase and immediately reflected on the passenger interface.

The synchronisation process occurred within a short time interval, demonstrating the real-time capabilities of Firebase.

<u>The following observations were recorded:</u>

- Location updates were successfully uploaded to the database.
- Passenger devices received updates automatically.
- No manual refresh was required.
- Synchronisation occurred across different devices connected to the internet.

The testing confirmed that Firebase effectively served as the communication bridge between drivers and passengers.

5.6 Cross-Device Communication

One of the primary objectives of RidePulse was to enable communication between different devices. This functionality was tested by using a smartphone as the driver device and a laptop as the passenger device.

The driver initiated live tracking through the Driver Dashboard, while the passenger accessed the Track Bus page from a separate device. As the driver's location changed, the passenger interface displayed updated location information almost immediately.

This experiment demonstrated that RidePulse successfully supports cross-device communication through cloud synchronisation.

The successful implementation of this feature significantly enhances the practical usability of the system.

5.7 Interactive Map Results

The Leaflet map component successfully displayed the live position of the bus using markers. Whenever new coordinates were received from Firebase, the marker automatically moved to the updated location.

The map provided the following benefits:

- Visual representation of bus location.
- Real-time movement updates.
- Improved user experience.

- Better understanding of route progression.

The integration of interactive maps significantly improved the usability of the application compared to displaying raw coordinates.

5.8 Challenges Faced During Development

Several challenges were encountered during the development and testing of RidePulse.

1. **Location Accuracy**

During early testing, location data was sometimes obtained through network-based positioning rather than GPS, resulting in reduced accuracy.

2. **Cross-Device Synchronisation**

Initially, local storage was used to store tracking information. However, local storage only functions within the same browser environment and cannot synchronise data across multiple devices.

This issue was resolved by integrating Firebase Realtime Database.

3. **User Interface Consistency**

Combining multiple pages and maintaining a consistent design across all modules required repeated testing and adjustments.

4. **Deployment Challenges**

After development, additional testing was required to ensure compatibility with Vercel deployment and cloud-based synchronisation.

Despite these challenges, all critical issues were successfully resolved before final deployment.

5.9 Discussion

The testing results indicate that RidePulse successfully achieves its primary objective of providing real-time bus tracking functionality.

The system demonstrated reliable performance across multiple devices and successfully synchronised location information through Firebase Realtime Database. The integration of GPS technology and interactive maps created a user-friendly environment for both drivers and passengers.

Although the current implementation focuses on a limited number of routes and buses, the architecture provides a solid foundation for future expansion. The use of cloud technologies ensures scalability and flexibility, making the system suitable for further development.

The project also demonstrates how modern web technologies can be utilised to solve practical transportation problems without requiring expensive infrastructure.

5.10 Chapter Summary

This chapter presented the results obtained during testing and evaluation of the RidePulse system. Functional testing, GPS tracking, Firebase synchronisation, cross-device communication, and map integration were successfully validated. The results confirm that RidePulse provides an effective solution for real-time bus tracking and achieves the objectives established during the project planning phase. The next chapter presents the overall conclusion of the project and discusses potential future enhancements.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The rapid growth of urban populations and increasing dependence on public transportation have created a need for efficient and reliable transport management systems. One of the most common challenges faced by passengers is the lack of real-time information regarding the location and movement of buses. This often results in uncertainty, inconvenience, and unnecessary waiting time.

RidePulse was developed as a web-based real-time bus tracking system to address these challenges. The project successfully demonstrates how modern web technologies, cloud databases, and GPS services can be integrated to create a practical transportation solution. The system enables drivers to share their live location through a dedicated driver interface while allowing passengers to monitor bus movement through an interactive map-based tracking platform.

The project was implemented using HTML, CSS, and JavaScript for frontend development, Firebase Realtime Database for cloud synchronisation, Leaflet Maps for geographical visualisation, and browser-based Geolocation APIs for location tracking. These technologies were combined to provide a simple, responsive, and user-friendly application capable of operating across multiple devices.

The testing and evaluation results confirmed that the system successfully achieved its primary objectives. Driver registration and login functionalities operated correctly, GPS coordinates were accurately captured, Firebase synchronisation enabled real-time communication between devices, and passengers were able to monitor bus locations through the tracking interface. The successful deployment of the project on Vercel further demonstrated the practical usability of the application in a real-world environment.

The project also provided valuable learning experiences related to web development, cloud computing, database integration, geolocation services, responsive design, and deployment technologies. Through the development process, various technical challenges were encountered and resolved, contributing significantly to the understanding of software engineering practices and problem-solving techniques.

Overall, RidePulse demonstrates that an affordable and accessible real-time transportation tracking system can be developed using widely available technologies. The project successfully fulfils its intended purpose and provides a strong foundation for future improvements and expansion.

6.2 Future Scope

Although RidePulse successfully achieves the objectives defined for the current implementation, there are several opportunities for future enhancement and expansion.

One potential improvement is the support for multiple buses operating simultaneously. The current system focuses on tracking a limited number of buses, whereas a future version could manage an entire fleet of vehicles through a centralised database.

Another valuable enhancement would be the implementation of Estimated Time of Arrival (ETA) prediction. By combining real-time location data with route information and traffic conditions, the system could provide passengers with more accurate arrival forecasts.

The development of a dedicated mobile application for Android and iOS platforms would further improve accessibility and user convenience. Mobile applications could provide additional features such as push notifications and offline route information.

Future versions of the system may also include an administrative dashboard for transportation authorities. Such a dashboard could provide route management capabilities, driver monitoring features, system analytics, and operational reporting tools.

Additional enhancements may include:

- Multi-bus and multi-route management.
- Estimated arrival time prediction.
- Passenger notification system.
- Mobile application development.
- Administrative control panel.
- Route analytics and reporting.
- Integration with traffic monitoring systems.
- Passenger feedback and complaint management.
- QR-code based bus identification.
- Smart ticketing integration.

With these enhancements, RidePulse can evolve from a prototype transportation tracking application into a comprehensive smart transportation management platform capable of serving a larger number of users and routes.

The project highlights the potential of combining web technologies, cloud computing, and geolocation services to improve public transportation systems. It also demonstrates how technology-driven solutions can contribute towards smarter, more efficient, and more user-friendly urban mobility services.

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APPENDIX

Project Title: Ride Pulse – A Real-Time Bus Tracking System

Development Technologies:

1. HTML
2. CSS
3. JavaScript
4. Firebase Realtime Database
5. Leaflet Maps
6. Geolocation API

Deployment Platform:

1. Vercel

Target Platform:

1. Web Browser (Desktop and Mobile)

Project URL:

https://ridepulse-two.vercel.app/

The appendix provides supplementary information related to the technologies and deployment environment used during the development of the Ride Pulse system.

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